Matrix functions and stability

Critical exponent for generalized doubly nonnegative matrices

Difficulty: challenging **Importance:** interesting to the community

Topic: Positivity of conventional matrix powers.

Problem statement

For $n \geq 3$, let $\mathcal{G}_n$ consist of the matrices $A \in \mathbb{R}^{n\times n}$ that are entrywise nonnegative, diagonalizable, and have only nonnegative real eigenvalues. Symmetry is not assumed. If

$$A = S\,\mathrm{diag}(\lambda_1, \dots, \lambda_n)S^{-1},$$

define the conventional matrix power, for $t > 0$, by

$$A^t = S\,\mathrm{diag}(\lambda_1^t, \dots, \lambda_n^t)S^{-1}, \qquad 0^t = 0.$$

Let

$$\mathrm{CE}_n = \inf\{c \geq 0 : A^t \text{ is entrywise nonnegative for every } A \in \mathcal{G}_n \text{ and every real } t > c\}.$$

Is $\mathrm{CE}_n = 2(n-2)$ for every integer $n \geq 3$?

Why it matters

This asks for the exact dimension-dependent threshold beyond which spectral matrix powers preserve entrywise positivity in a nonsymmetric class, relevant to matrix functions and positive dynamics.

References

- X. Han, C. R. Johnson, and P. Paparella, The critical exponent for generalized doubly nonnegative matrices, *Linear and Multilinear Algebra* 65 (2017), 1035–1044, §1 for definitions, Theorem 3.3 for an upper bound, Corollary 4.5 for $n = 3$, and Question 4.7 for the displayed formula. The conjectural $n = 4$ case is Conjecture 4.6.
- D. Guillot, A. Khare, and B. Rajaratnam, The critical exponent conjecture for powers of doubly nonnegative matrices, *Linear Algebra and its Applications* 439 (2013), 2422–2427, abstract and main theorem: resolution of the symmetric case.

Status check

On 2026-09-08, checked the latest listed arXiv version and searched “generalized doubly nonnegative”, “critical exponent”, “2(n-2)”, “conjecture”, and 2025–2026. No resolution of Question 4.7 was located. The source proves existence of a finite threshold and $\mathrm{CE}_3 = 2$. The solved symmetric doubly nonnegative problem has threshold $n-2$ and concerns a smaller class; results for entrywise (Hadamard) powers concern a different operation. The special case $n = 4$ and related integrality questions are included within this single target, not counted separately. No recent explicit reaffirmation was found.